Olympics Analytics: A Data-Driven Study of Global Performance, Participation, and Trends (1896–2016)

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Problem Statement

The Olympic Games represent one of the largest global sporting events, involving athletes from numerous countries across a wide range of sports. However, analyzing long-term trends such as participation growth, gender representation, country dominance, and sport-wise performance is challenging due to the scale and complexity of the data.

This project aims to answer key questions such as:

- How has athlete participation evolved over time?
- How has gender representation changed across decades?
- Which countries dominate the Olympics, and how many athletes do they send?
- Which sports contribute the most to medal distribution?
- What factors influence athlete performance and success?

The objective is to transform raw Olympic data into meaningful insights using interactive visualizations.

Features

This project presents a Power BI report consisting of multiple pages, each focusing on a specific analytical aspect and presented with relevant slicers:

Page 1: *Dashboard*

- Key KPIs:
    1) Total Athletes
    2) Total Countries
    3) Total Sports
    4) Total Country Medals

- Athlete participation trend over time
- Top 10 countries by medal count

Page 2: *Participation & Gender Analysis*

- Male vs Female participation trend
- Gender ratio evolution (percentage-based visualization)
- First Year of Female participation across various sports

Page 3: *Country Performance & Dominance*

- Athlete participation vs Medal count trend
- Medal type breakdown of top-performing countries
- Medal trends of top countries over time

Page 4: *Sports & Events Analysis*

- Medal distribution across sports (Treemap)
- Number of events per sport
- Total events vs Medals trend

Page 5: *Athlete Insights*

- BMI vs Medals scatter plot (sport-wise success analysis)
- Age distribution of athletes
- Growth in success with experience trend

Page 6: *Advanced Insights & Conclusions*

- Medal efficiency (medals per athlete)
- Sport-wise Country dominance
- Host country advantage analysis

Screenshots

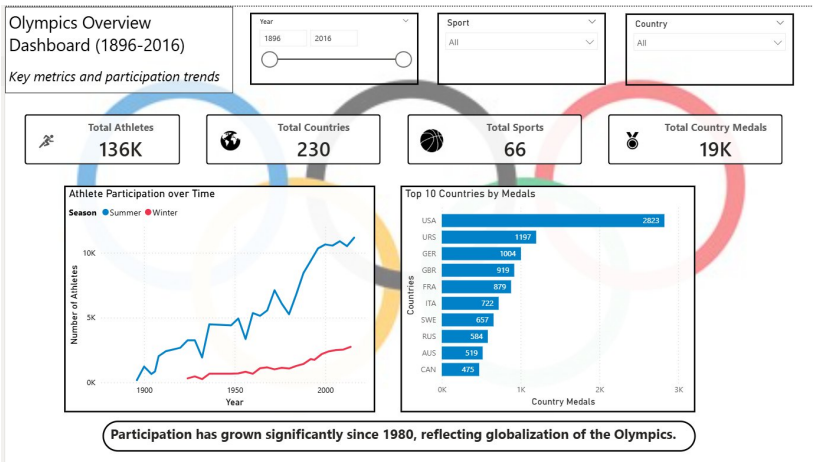

Figure 1: Dashboard

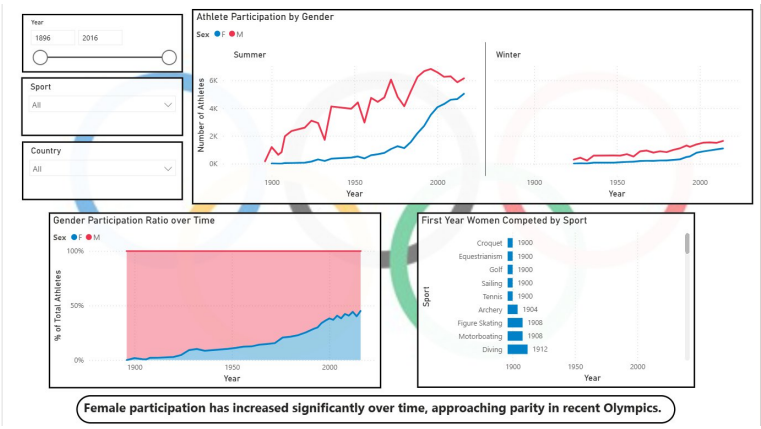

Figure 2: Participation & Gender Analysis

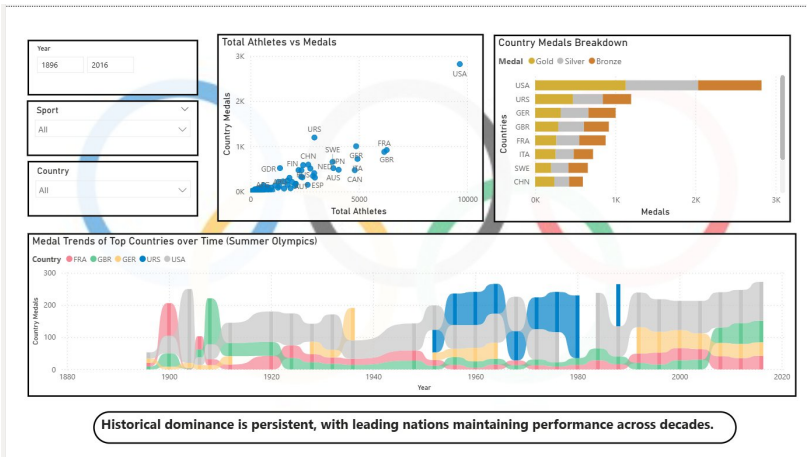

Figure 3: Country Performance & Dominance

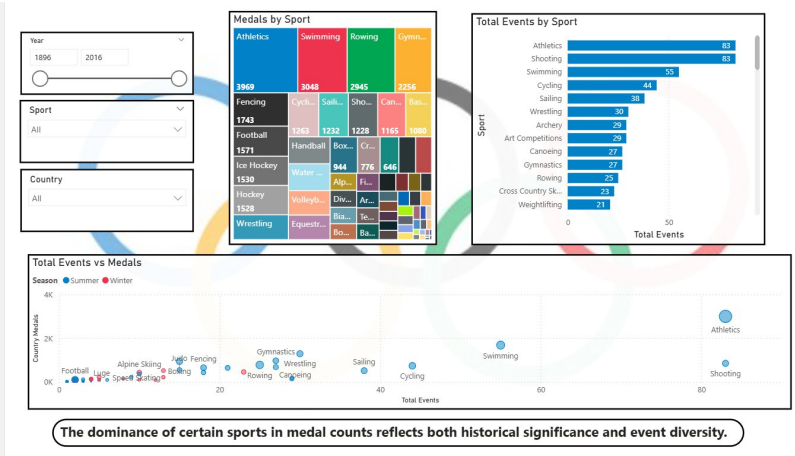

Figure 4: Sports & Events Analysis

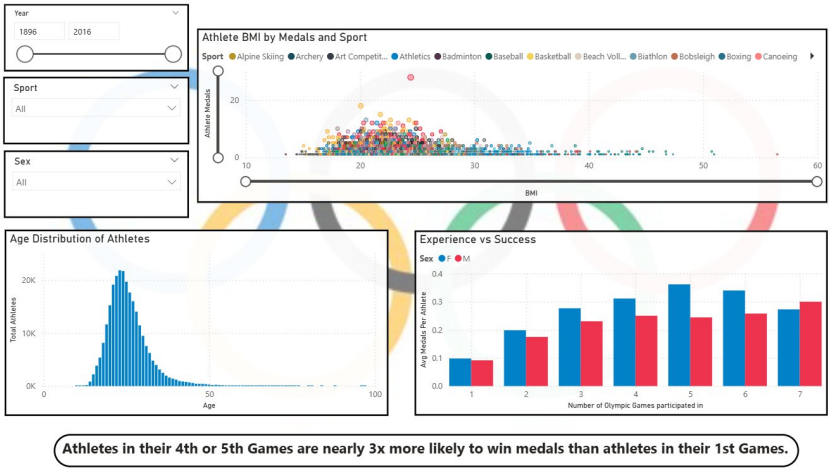

Figure 5: Athlete Insights

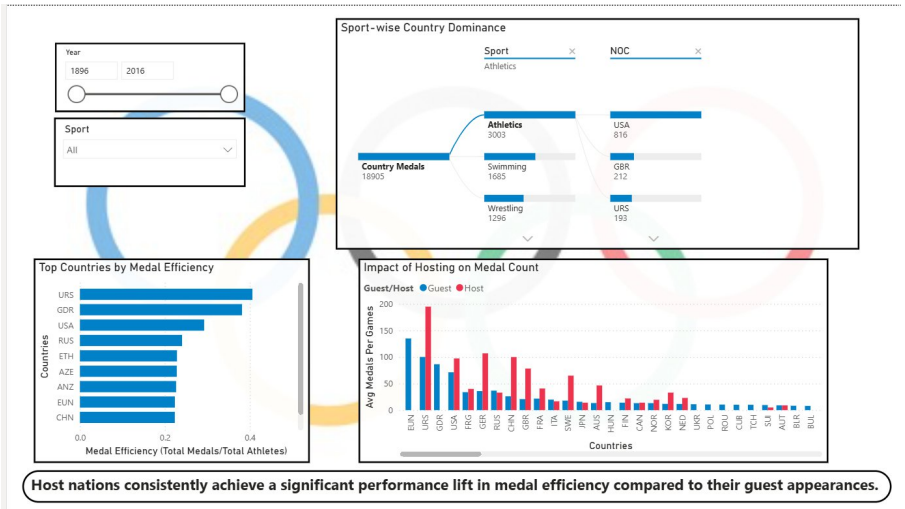

Figure 6: Advanced Insights & Conclusions

Tech Stack

- Power BI – Data transformation and report creation
- CSV – Data source (Kaggle link)
- JSON – Olympic theme colour

Unique Points

- Comprehensive analysis covering 120 years of Olympic history
- Multi-dimensional insights:
    - Temporal trends
    - Gender evolution
    - Country dominance
    - Sport specialization
- Use of advanced metrics such as:
    - Medal Efficiency (Medals per Athlete)
    - Gender participation ratio
- Interactive dashboard with slicers for dynamic exploration

Future Improvements

- Integration of real-time Olympic data APIs
- Inclusion of predictive analytics (e.g., medal prediction models)
- Addition of machine learning models for performance analysis
- Enhanced geographic analysis using more detailed location data
- Development of a web-based interactive dashboard for public use